



Faculty of Computing and AI

CS213 Object Oriented Programming
Year 2026-2027
First Semester

Assignment 1 — Version 8.2

Course Instructors:
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Revision History

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Version 2.0	By Dr Mohammed El-Ramly	2 April 2018	Bonus added
Version 3.0	By Dr Mohammed El-Ramly	9 April 2018	Added more filters
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Objectives

This assignment reviews CS112 Structured Programming concepts and trains students on problem solving using C++ arrays and ready-made library code, written by others for us to use. Students will work in teams to build a real image processing system and learn about that domain.

Preparation

1. Review C++ syntax and refresh your C++ knowledge.
2. Read / watch any quick tutorial on C++.
3. Create a **private GitHub** repo for the project and **use it for development**.

Introduction

Computer vision / image processing algorithms are behind a wide variety of technologies that we use in our everyday lives. Visit the page <https://ai.stanford.edu/~syyeong/cvweb/applications.html> to get an overview of a few applications for this field. Examples are: **image filtering**, which is an important part of photo editors and applications like Photoshop or Instagram; **image matching**, which is used for retrieval and stitching together panoramas; **image segmentation**, which is implemented in the technology behind green screens in filmmaking industry and **image recognition**, which is required for products like Google image search and Facebook automatic face tagging.

In this assignment, you will develop an image processing tool that can apply different filters (changes) to a given image of any size and the four popular image formats. You are given:

- (1) A small library with four functions for loading the pixels of the image into an array . The library has two files **stb_image.h** and **stb_image_write.h** and a file for class **image** called **Image_Class**. You **only** need to include this line at the top of your program to include the library with your code. Put these files in the same directory as your code.

```
#include "Image_Class.h"
```

- (2) A set of images to work on (but you can bring your own) and an sample small program.
- (3) A description of the filters required.
- (4) A playlist to show how to use the library:

<https://www.youtube.com/playlist?list=PLT8-zF1fr5DiXUgC7GXaFWnMVgdoAFLZM>

- (5) A FAQ page answers many of the questions you may have.

<https://docs.google.com/document/d/1zZq50DO0oFrj9NsFo0dEXAQX66dnLQB5-OtRz3xLNE0/edit?usp=sharing>

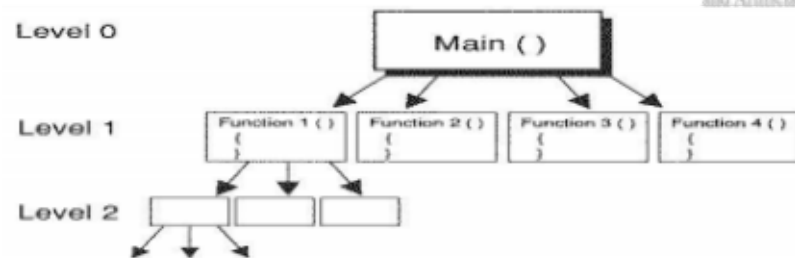
You will develop a program that displays a menu of choices for the user continuously until the user chooses exit. The menu will allow the user to load an image and choose the filter to apply. The user can choose (save) to store the image resulting from applying the filter after specifying the name of the target file. The program will loop until the user chooses exit.

For your program, you should develop each filter in a separate function. Deliver a **system diagram** showing the **different functions** of the system and **their relation** to each other as shown in the figure below. Put this diagram in a shared doc and put the link in the code.



Sample System Diagram

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Instructions

- It is very important to collect coursework marks so that you pass easily and get a good grade.
من المهم للغاية حسن أداء أعمال السنة لتتجح بسهولة و تحصل على تقدير مرتفع
- These instructions must be followed to get the full marks.
- The assignment includes two levels (norma and bonus):
 - Hero level - where students implement 16 filters مستوى البطل
 - Beast level - where student will do 2 extra filters and GUI مستوى الوحش
- Part 1 Due Date: 1 Oct 2026 (Deliver ANY 8 filters + a menu. Any 8 are OK)
- Part 2 Due Date: 11 Oct 2026 (For hero, deliver 16 filters. For Beast, deliver GUI and 18 filters)
- Students will form teams of four students from the same lab or section.
- AI policy. For Tasks 1,2,3 (16 filters), no use of AI or code generation tools is allowed. Any generated or copied code gets zero for all team. For Task 4 (Filters 17 & 18 and GUI), using AI is permitted. BUT students must fully understand and explain the code and be able to rewrite it, or fix it or explain it.**

Task 0 – Basics of Image Representation

Images in computer systems are represented as a matrix of pixel values. Each pixel holds numbers related to the intensity of the composition of Red, Green, and Blue (RGB values). These values range from 0-255 providing a wide range of color variations. If Red is 0 it means that there is no Red component in the image

Applying image filters involves processing each pixel in an image according to a specific algorithm or set of rules. These algorithms manipulate the pixel values to achieve desired effects such as blurring, sharpening, adjusting colors, and adding artistic effects.

Assume a 3x3 Image, each pixel (cell) contains 3 channels R G B, for example:

Red 255, Green 255, Blue 255 = White	0, 0, 0 = Black	255, 0, 0 = Red
124, 124, 124 Gray scale (equal channels)	255, 0, 255 = Pink	255, 255, 0 = Yellow
0, 255, 255 = Bright blue		X, Y, Z is some other color

Watch more: <https://youtu.be/pmY7pOQCOr8?list=PL6p1q8upS1ABS7L--0dl62PvKSNIJHkrU>
<https://www.youtube.com/watch?v=IZXemaxIxJU>

Applications: <https://www.youtube.com/watch?v=aDpnaxPAmtU>
<https://medium.com/@zaheeribrahim346/uses-of-image-processing-840288728703>



Task 1 – Menu and File Loading and Saving (Group Task) - 1 mark

- Each team will collaboratively develop the menu display function, as well as the load and save functions. These functions should support saving images in any format, including JPEG, JPG, PNG, and BMP, and give the user freedom of naming the image. Team members will work on their respective tasks and then integrate and test all program functions together to be called from the menu. Failure to deliver an integrated program will result in a deduction of marks.
- The menu will allow the user to load a new image, apply a filter on the current image or store the image in a new or same file. The menu will loop forever till the user exits.
- The menu will work as follows:
 1. At the start of the program ask the user to load a file to work on. This becomes the current image.
 2. Then the program checks that this file exists and has a suitable extension
 3. Then the program will display a menu of options like this:
 - Load a new image
 - Filter 1 // name the filter
 - Filter 2 //
 -
 - Save the image
 - Exit
 4. After applying any filter, the same menu comes again
 5. Some menu options have other sub-options. For example, framing filter will ask if s/he wants a simple or fancy frame and will ask about the color.
 6. The current image becomes the result of the last filter
 7. This can be the same dimensions or a new image in cases of merge, resize, rotate, etc.
 8. The user can load a new current image to work on.
 9. At any point, s/he can save the current image.
 10. S/he is asked if he wants to save on the same file or change file name.
 11. The saving function verifies that the extension is a valid one. (.jpg, .bmp, etc.)
 12. At any point he can load a new current image and s/he is asked if s/he wants to save the current one before loading a new one.
 13. At any point s/he can exit and s/he is asked if s/he wants to save before exit.



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Task 2 – Using Github (Group Task) - 1 mark

- The team will create a **private** repo (Not public so no one steals your code) on GitHub and maintain all needed files there **and do their development through the repository** via work cycles of **init, clone, change, add, commit, remote add origin, push**
- The team will demo their repo to the TA.

Task 3 – Baby Photoshop for Image Processing (Individual/Group Task) - 5 marks

Each team member will develop 4 individual filters. The member with the smallest ID will do filters 1, 5, 9, 13 and the member with the next ID will do 2, 6, 10, 14 and the member with the next ID will do 3, 7, 11, 15 and one with largest ID will do 4, 8, 12, 16. **All team** are responsible for integrating all filters and testing them and ensuring they work properly together. Team should find more images to test on.

An image processing (or photo editing) software like PhotoShop allows you to load an image and apply some changes (called filters) to it and then save it again. **Some filters will apply on the same image and some will result in a new image to store (with the same name or a new one) or ignore it.**

يقوم برنامج معالجة الصور مثل الفوتوشوب مثلاً بتحميل الصورة المرغوبة وإتاحة مجموعة من العمليات تسمى المرشحات للتنفيذ على هذه الصور ، كل منها ينتج عمل تغيير في الصورة ، وفيما يلي شرح للمرشحات المطلوبة و في أعلاه شرح كيف سنقسم على فريق العمل و ما نصيب كل عضو، بعض الفلاتر ستطبق على نفس الصورة وبعضها سيعمل صورة جديدة بأبعاد جديدة يجب أن تسأل المستخدم هل يريد تخزينها أم لا و هل بنفس الاسم او اسم جديد.

Filter 1: Grayscale Conversion

A grayscale image is an image where each pixel is represented by a single Sample indicating the intensity of light or shade. In grayscale images, there is no color information, only varying shades of gray from black to white. The intensity value of each pixel determines its brightness, with lower values representing darker shades and higher values representing lighter shades.

صورة بدرجات الرمادي هي صورة يتم فيها تمثيل كل بكسل بعينة واحدة عادةً ، تشير إلى شدة الضوء أو الظل. في الصور بدرجات الرمادي، لا توجد معلومات عن الألوان، بل هناك فقط درجات متفاوتة من الرمادي من الأسود إلى الأبيض. تحدد قيمة الشدة لكل بكسل سطوعه، حيث تُمثل القيم الأقل قدرات أظلم والقيم الأعلى قدرات أفتح

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Filter 2: Black and White

Black and white images also lack color, but they typically have only two colors: black and white (0 and 255). There are no intermediate shades of gray in a true black and white image. Each pixel is either black (representing darkness or absence of information) or white (representing brightness or presence of information). In digital imaging, black and white images are often created by converting color images to grayscale and then applying a threshold to convert shades of gray to either black or white.

الصور الأبيض والأسود تفتقر أيضًا إلى الألوان، ولكنها عادة ما تحتوي على لونين فقط: الأسود والأبيض (0 و 255). لا توجد درجات رمادية وسطى في الصورة الأبيض والأسود الحقيقية. كل بكسل إما أن يكون أسودًا (ممثلاً للظلمة أو عدم الوجود للمعلومات) أو أبيضًا (ممثلاً للسطوع أو وجود المعلومات). في التصوير الرقمي، غالبًا ما يتم إنشاء الصور الأبيض والأسود عن طريق تحويل الصور الملونة إلى درجات الرمادي ثم تطبيق حد معين لتحويل درجات الرمادي إما إلى الأسود أو الأبيض.





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Filter 3: Invert Image

Inverting colors is a simple yet powerful image processing operation that transforms the colors of an image by reversing each pixel's RGB values. This operation effectively produces a photographic negative effect, where light areas become dark and vice versa.

تعتبر عملية عكس الألوان عملية معالجة صور بسيطة ولكنها قوية تقوم بتحويل ألوان الصورة عن طريق عكس قيم الاحمر والاخضر والازرق لكل بكسل. تؤدي هذه العملية بفعالية إلى إنتاج تأثير سلبي فوتوغرافي، حيث تصبح المناطق الساطعة مظلمة والعكس صحيح

Design and implement an algorithm for inverting image colors.



Filter 4: Adding a Frame to the Picture

When we hang photos on the wall, we put them in a frame. This filter adds a frame (a simple one) أضيف إطاراً للصورة إما إطار بسيط أو إطار مزخرف حسب خيار المستخدم





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Filter 5: Flip Image

Flipping an image involves mirroring its contents horizontally or vertically. Horizontal Flip: Also known as a left-right flip, this operation reflects the image along a vertical axis. Vertical Flip: Also known as a top-bottom flip, this operation reflects the image along a horizontal axis.

قلب الصورة يتضمن عكس محتوياتها أفقياً أو عمودياً القلب الأفقي: المعروف أيضاً بالقلب اليمين-اليسار، هذه العملية تعكس الصورة على محور رأسي القلب العمودي: المعروف أيضاً بالقلب الأعلى-السفلي، هذه العملية تعكس الصورة على محور أفقي

Develop an algorithm for applying Flipping effects to images.



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Flipped Vertical



Flipped Horizontally

Filter 6: Rotate Image

This filter allows the user to rotate the image clockwise by 90°, 180° or 270° as the user chooses.

مرشح الدوران: هذا المرشح يدير الصورة باتجاه دوران الساعة 90 أو 180 أو 270 درجة



Rotate 90 degrees



Rotate 180 degrees



Rotate 270 degrees



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Filter 7: Darken and Lighten Image (Two filters in this task)

This filter allows the user to make the image darker or lighter by 50% from the original image. Or better, allow the user to choose the level of darkening or lighting from 0 to 100%. Use a slider or a suitable GUI element.

مرشح الإضاءة : يتيح هذا المرشح تفتيح و تغميق الصورة الأصلية بنسبة 50% بمعنى زيادة أو تقليل إضاءة الصور ب 50%.



Filter 8: Resizing Images

Resizing an image involves changing the dimensions of the entire image while maintaining the same image details. User can enter new dimensions or can enter a ratio of reduction or increase.

تغيير حجم الصورة: ينطوي تغيير حجم الصورة على تغيير أبعاد الصورة بأكملها مع الحفاظ على نفس تفاصيل الصورة

Original image: 1000 x 700



Resized to 500x500 :



Resized to 1280 x 800:





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Filter 9: Merge Images

Merging two images typically involves combining them in some way to create a single composite image. If the two images are the same size, then we only need to merge them directly. However, if the images are of different sizes, we need to allow two options: (1) resize the smaller image or both images to the biggest height and the biggest width and then merge and (2) the other option is to merge the common area of the smaller width and the smaller height. You must allow both. For grand competition, allow dragging one picture on the other to show which area will overlap.

دمج صورتين عادةً ما ينطوي على دمجهما بطريقة ما لإنشاء صورة مركبة واحدة جديدة. إذا كانت الصورتان بنفس الأبعاد، فإننا نحتاج فقط لدمجهما مباشرة. ومع ذلك، إذا كانت الصور ذات أحجام مختلفة، فإننا علينا تغيير حجم الصورة الأصغر أو الاثنين باستعمال فلتر تغيير الحجم، أو قص الصورة الأكبر بحيث تندمج الصورة الصغرى مع الجزء المماثل من الكبرى ويمكن أيضاً لو حجمهما مختلف دمج المساحة المشتركة منهما، يعني مثلاً صورة 500 في 1000 و أخرى 1000 في 500 سيتم عمل صورة جديدة 500 في 500 تشمل الجزء المشترك منهما.

Create a Function for merging two images.





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Filter 10: Detect Image Edges

Edge detection is a fundamental technique in image processing used to identify and highlight the boundaries of objects within an image. The goal of edge detection is to find areas in an image where the intensity or color changes abruptly, indicating the presence of an edge or a boundary between different objects or regions.

الكشف عن الحواف هو تقنية أساسية في معالجة الصور تُستخدم لتحديد وإبراز حدود الأجسام داخل الصورة. الهدف من كشف الحواف هو العثور على المناطق في الصورة حيث يتغير التباين أو اللون بشكل حاد، مما يشير إلى وجود حافة أو حدود بين أجسام أو مناطق مختلفة (قد يكون من الأسهل تحويل الصورة لأبيض و أسود ثم اكتشاف الحواف عليها)



Filter 11: Crop Images

Cropping: Cropping an image involves removing portions of the image from one or more sides, effectively changing its composition. Unlike resizing, cropping does not necessarily preserve the original details

التقطيع: عملية تقطيع الصورة تتضمن إزالة جزء من الصورة من جهة أو أكثر، مما يغير تكوينها بشكل فعال. على عكس تغيير الحجم، التقطيع لا يحافظ بالضرورة على التفاصيل الأصلية.

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Create an algorithm for cropping where the user can specify the dimensions to crop to, provided that the dimensions are smaller than or equal to the original image.

You will take x and y input as the starting point (upper left corner of the part to keep) and $W \times H$ as the dimensions of the area to cut. This area is cut and stored in a new image or with the same name if the user wants



cropped at **corner**(700,700) with dimensions 600x600

Filter 12: Blur Images

Blurring an image is a technique used in image processing to reduce the level of detail or sharpness in an image, resulting in a smoother appearance. **For grand competition, allow different levels of blurring.**

تقوم عملية تشويش الصورة بتقنية مستخدمة في معالجة الصور لتقليل مستوى التفاصيل أو الحدة في الصورة، مما ينتج عنه مظهرًا أكثر نعومة.

Develop an algorithm for applying blur effects to images.

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The Land of Wano

The Land of Wano is a mysterious and isolated country in the world of One Piece. It's inspired by feudal Japan, with samurai, ninja, and traditional Japanese culture prevalent throughout. Wano is known for its closed borders, keeping outsiders away and maintaining its own unique way of life.

أرض وانو هي بلد غامض ومعزول في عالم ون بيس. إنها مستوحاة من اليابان الفيودالية، مع وجود الساموراي والنينجا والثقافة اليابانية التقليدية في كل مكان. وانو مشهورة بحدودها المغلقة، حيث تبقى الأجانب بعيدًا وتحافظ على طريقة حياتها الفريدة

Filter 13: Wano does not have good natural sunlight. Can you fix that?

وانو لا تمتلك أشعة شمس طبيعية جيدة، هل يمكنك تصحيح ذلك؟

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Filter 14: Wano villagers use old Den Den Mushi (televisions) to view the news. Can you implement their TV images?

This is the effect you need to add to the image which is like how old TA screens used to add to images <https://www.shutterstock.com/image-vector/no-tv-signal-on-retro-televisor-516537094>

يستخدمون دن دن موشى القديمة لمشاهدة الأخبار. هل يمكنك تنفيذ صور التلفزيون الخاصة بهم؟





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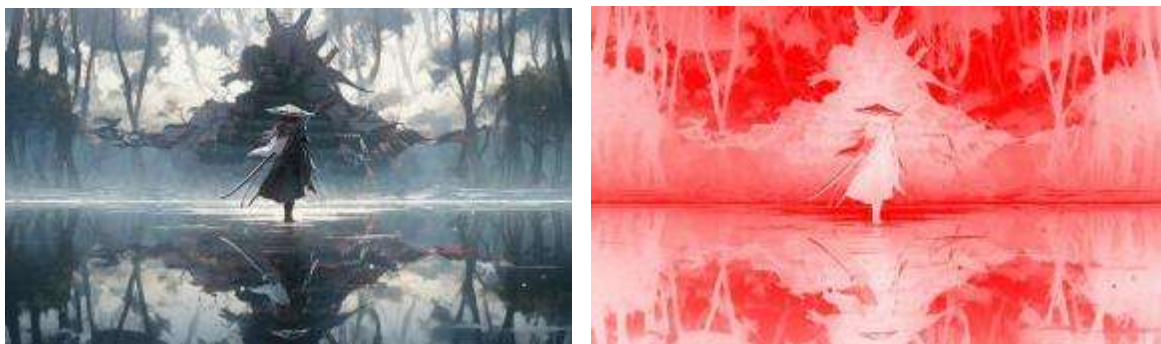
Filter 15: Wano looks so purple at night can you make Luffy look purple?

وانو تبدو باللون الأرجواني في الليل، هل يمكنك جعل لوفي باللون الأرجواني بشكل كافٍ؟



Filter 16: Samurais are passionate about capturing the world in infrared photography—let's empower their pursuit !

الساموراي يحبون التقاط الصور الأشعة تحت الحمراء، لنساعدهم في ذلك



Filter 17: Image Skewing Filter

A skew image filter is a digital image processing technique used to distort or skew an image. It can be employed for various purposes such as correcting perspective distortion, creating artistic effects, or simulating 3D perspectives.

فلتر التموج الصورة هو تقنية معالجة الصور الرقمية تُستخدم لتشويه أو تمويه الصورة. يمكن استخدامه لأغراض مختلفة مثل تصحيح تشويه النظرة، وخلق تأثيرات فنية، أو تحاكي مناظر ثلاثية الأبعاد.

Develop an algorithm for applying Skew effects to images as shown below. User can enter the degree of skewness needed along the vertical axis.

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Image skewed by 40 degrees

Filter 18: Oil painting is famous in Wano. Can you oil paint images?

الرسم بالزيت مشهور في وانو فهل يمكنك إثبات أنك ساموراي وترسم صوراً بالزيت؟

Notice how in Oil painted images the sky, water, and lights look why do you think that happened?

لاحظ كيف تبدو السماء والماء والأضواء في صور الرسم بالزيت، لماذا تعتقد أن ذلك حدث؟



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Task 4 – Filters 17 & 18 and GUI (Bonus Task) - 2 marks

1. Team will implement all filters mentioned in this document, including 17 & 18 & will implement a GUI to show the images and modifications as they happen. See some sample simple videos of work of students from previous years.
2. Team can use any GUI package like QT, Windows Forms, FLTK or wxWidgets or others.
3. Team can use AI generators ONLY for this task but will only get mark if fully understand and can rewrite the code.

Howto Submit

In all cases, submit only code. NO exe. DO NOT submit the library files unless asked.

Part 1 (8 filters) (1 Oct 2026)

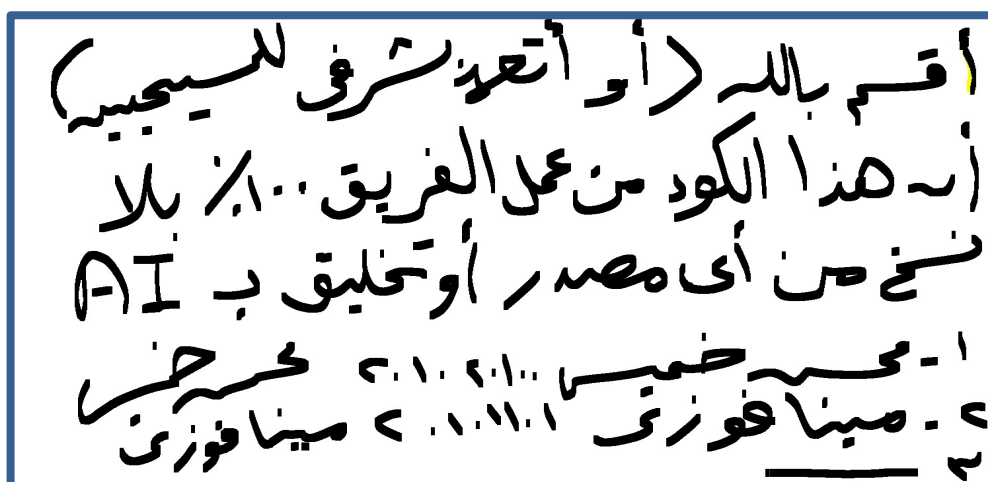
Submit **only** a .cpp file (No GUI, only console app) that has a header that explains the file and all team details and IDs and section and who did which filter. Then the file should have a menu that displays the options to choose from.

Name the file: CS213_A1_Part1_SectionNum_YourID_YourID_YourID_YourID.cpp.

Part 2 Hero Level (16 filters - Tasks 1,2,3) and Beast Level (18 filters+GUI, T. 1,2,3,4) (11 Oct 2026)

Do a **short video** (3-5 min) and load in a public place or drive that introduces the team members, professor, course name and number and year, and FCAI and its logo. Then, it explains how to compile and run the program and show the operation of different filters and their effect and other options in the program.

Create a **shared document** that has team details, description of **who did what**, snapshots from **GitHub repo**, and system **breakdown diagram** mentioned earlier in this document. Also, include snapshots from your program. Also, explain the **best and hardest parts in assignment**. **Shared doc should have a scan of a hand written oath like this:**



If not doing GUI, Submit a .cpp file that has a header that has the following:

- a) All team details and IDs and section
- b) Work breakdown - details of who did which filter and who did the menu
- c) Link to the video demo of the program.
- d) Link to the shared document.

Name the file: **CS213_A1_Part2_SectionNum_YourID_YourID_YourID_YourID.cpp**.

If you are doing GUI, Submit a zip file with

1. All code files you developed and the header files of the library files given to you.
2. The project file you created that contains the code files.
3. Do not include any additional files or libraries like Qt or others. **Do NOT include exe files.**
4. Main .cpp file that should have a header that explains the file and all team details and IDs and section and who did which filter. Then the file should have a menu that displays the options to choose from saving, loading and the available filters.
5. A .txt read me file that explains in details what is needed to set up and run the code and what tools and versions of these tools you used.
6. A .txt or .html link to a video showing your program working (not the code) and different options in it and sample output for every filter. Video should have FCAI logo and date and names of authors and should explain that this is done as part of CS213 OOP Programming at FCAI, Cairo University under supervision of Dr Mohammad El-Ramly. Also a link to the shared doc mentioned in part 2.

Name the file: **CS213_A1_Part2_SecNum_YourID_YourID_YourID_YourID.zip**

Marking criteria

- | | |
|----------|---|
| 1 mark | evidence that the team used GitHub and had a repo for the project (all team, loaded multiple times on GitHub) |
| 1 mark | program is integrated, has a united menu and can load and save files (whole team) |
| 4 marks | individual filters are implemented and working (individual) |
| 1 mark | group finished all the filters and good teamwork (whole team) - if some missing, deduct |
| -1 mark | no video explaining the program and showing the filters (whole team) |
| -1 mark | no shared doc with team info, screenshots, system diagram, breakdown, etc. (team) |
| 2 marks | Bonus - for doing (18 filters + GUI) (only members who participated in it) |
| -7 marks | For whole team, if any part of the assignment is copied, generated or u cannot explain. |